

- 3 (a) Fig. 3.1 is a photomicrograph showing part of a transverse section through the root of an iris, *Iris germanica*. Irises are herbaceous monocotyledons. These plants have the same transport tissues as herbaceous dicotyledons, but the transport tissues are distributed differently. In monocotyledons, the central tissue in the root is parenchyma (packing tissue).

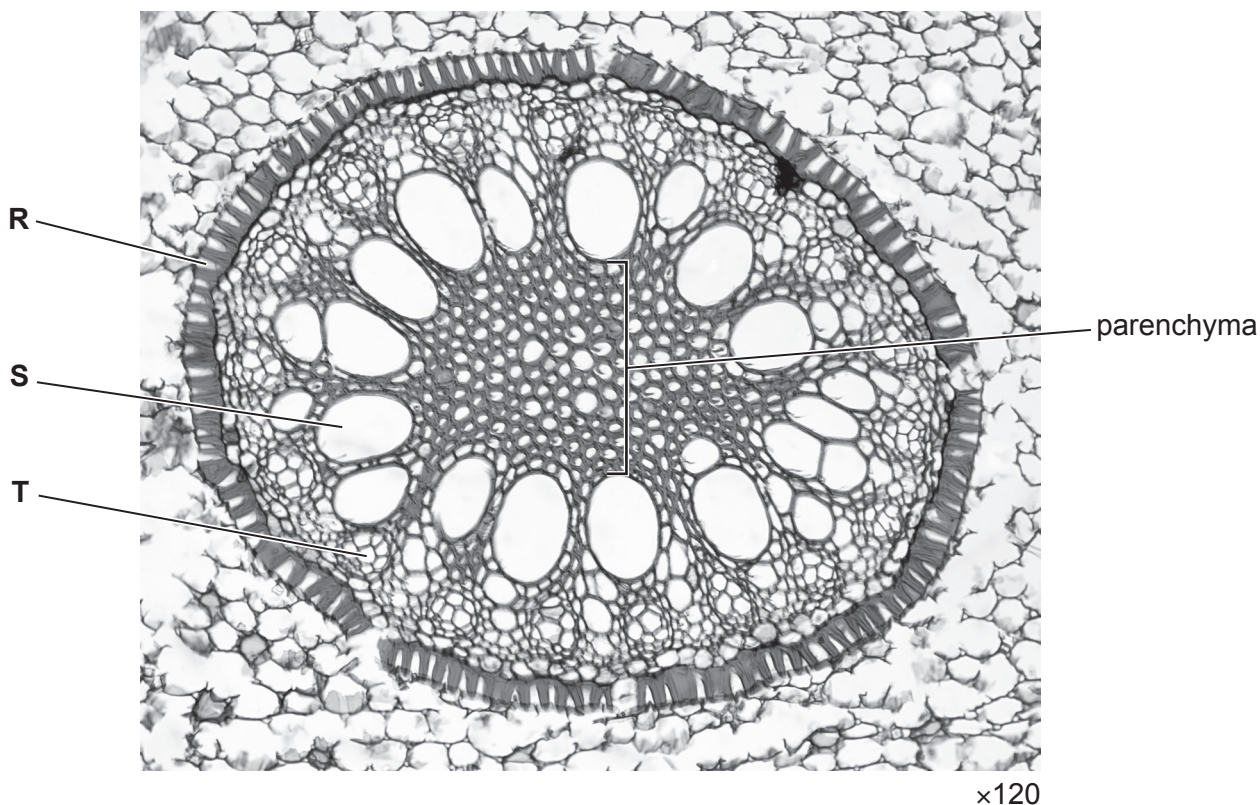


Fig. 3.1

- (i) Cells **R**, **S** and **T** in Fig. 3.1 are found in different tissues.

Name the tissues in which the cells labelled **R**, **S** and **T** are found.

tissue in which cell **R** is found

tissue in which cell **S** is found

tissue in which cell **T** is found

[3]

- (ii) Outline the role of the tissue in which cell **R** is found in Fig. 3.1.

.....

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.....

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..... [2]

(iii) State an example of an organic compound that is translocated in the root of an iris.

..... [1]

(b) The electron micrograph in Fig. 3.2 shows a section through some root cells in an onion, *Allium cepa*.



×6250

Fig. 3.2

On Fig. 3.2, draw a label line and label it with the letter **P** to identify **one** plasmodesma. [1]

(c) Table 3.1 contains information about four polysaccharides found in animals or plants.

Complete Table 3.1 by filling in the missing information.

Table 3.1

polysaccharide	monomer	glycosidic bond(s)	function
amylopectin	α -glucose	1,4 and 1,6	energy storage in plants
amylose		1,4	energy storage in plants
cellulose	β -glucose		structural role in plant cell walls
glycogen	α -glucose	1,4 and 1,6	

[3]

[Total: 10]

Question 4 starts on page 14.